Linux OS and Scripting

LAB Assignment 5

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Subject: Linux OS and Scripting

Part A: Authentication Log

Exercise 1: To print, Write sed commands to:

Print the first log entry:

sed -n '1p' auth.log

Print the fifth log entry:

sed -n '5p' auth.log

Print entries from lines 2 to 6:

sed -n '2,6p' auth.log

Print the last entry:

sed-n'$p'auth.log

Print every second log entry starting from line 1:

sed -n '1~2p' auth.log

Print all entries containing Failed password:

sed -n '/Failed password/p' auth.log

Print all successful login entries containing Accepted password:

sed -n '/Accepted password/p' auth.log

Print entries from the first failed login to the authentication failure:

sed -n '/Failed password/, /authentication failure/p' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Failed password/' auth.log
sed: -e expression #1, char 17: missing command
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$
```

Exercise 2: Filtering Security Events

Write commands to:

Display failed login attempts:

sed-n'/Failedpassword/p' auth.log

Display successful login attempts:

sed-n'/Acceptedpassword/p'auth.log

Display entries associated with the root account:

sed -n '/root/p' auth.log

Display entries containing Invalid user:

sed -n '/Invalid user/p' auth.log

Display all events related to IP address 172.16.0.8:

sed -n '/172\.16\.0\.8/p' auth.log

Display all events that do not contain Accepted password:

sed -n '/Accepted password/!p' auth.log

Display all events except those generated by sudo:

sed -n '/sudo/!p' auth.log

Display only the entries that contain either Failed password or Invalid user using multiple -e commands:

sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/172\.16\.0\.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n -e '/Failed password/p' -e '/Invalid user/p'  auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$
```

Exercise 3: Removing Irrelevant Entries

Without modifying the original file, display the log after:

Deleting successful login entries.

sed '/Accepted password/d' auth.log

Deleting entries containing Connection closed.

sed '/Connection closed/d' auth.log

Deleting the first three entries.

sed '1,3d' auth.log

Deleting entries from lines 4 to 7.

sed '4,7d' auth.log

Deleting all entries associated with 192.168.1.15.

sed '/192\.168\.1\.15/d' auth.log

Deleting every second entry starting from the first line.

sed '1~2d' auth.log

Deleting all events that do not contain Failed password.

sed '/Failed password/!d' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/192\.168\.1\.5/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/192\.168\.1\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
```

Exercise 4: Classifying Security Events

Replace Accepted password with [SUCCESSFUL LOGIN].

sed's/Acceptedpassword/[SUCCESSFULLOGIN]/g'auth.log

Replace Failed password with [FAILED LOGIN].

sed's/Failedpassword/[FAILEDLOGIN]/g'auth.log

Replace Invalid user with [INVALID USER].

sed's/Invaliduser/[INVALIDUSER]/g'auth.log

Replace authentication failure with [AUTHENTICATION FAILURE].

sed's/authenticationfailure/[AUTHENTICATIONFAILURE]/g' auth.log

Replace root with PRIVILEGED_USER.

sed's/\broot\b/PRIVILEGED_USER/g' auth.log

Replace every occurrence of server with authentication-server.

sed 's/server/authentication-server/g' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/Failed password/[FAILED LOGIN]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/Invalid user/[INVALID USER]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/\broot\b/PRIVILEGED_USER/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
```

Exercise 5: Masking Sensitive Information

Use sed to anonymise the log before sharing it.

Replace IP address 10.0.0.25 with X.X.X.X.

sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log

Replace 172.16.0.8 with SUSPICIOUS_IP.

sed's/172\.16\.0\.8/SUSPICIOUS_IP/g'auth.log

Replace username admin with USER_A.

sed's/\badmin\b/USER_A/g'auth.log

Replace username student with USER_B.

sed's/\bstudent\b/USER_B/g'auth.log

Replace all digits with X.

sed's/[0-9]/X/g'auth.log

Replace periods in IP addresses with hyphens.

sed's/\([0-9]*\)\.\([0-9]*\)\.\([0-9]*\)\.\([0-9]*\)/\1-\2-\3-\4/g' auth.log

Replace all IP addresses in the sample file with [REDACTED_IP].

sed's/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.X
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/\badmin\b/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/\bstudent\b/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/\([0-9]*\)\.\([0-9]*\)\.\([0-9]*\)\.\([0-9]*\)/\1-\2-\3-\4/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

Part B: Web-Server Log

Use access.log to solve the below questions

Write sed commands to:

Print all requests that returned status code 200.

sed -n '/200/p' access.log

Print all requests that returned status code 401.

sed -n '/401/p' access.log

Print all requests that returned status code 403.

sed -n '/403/p' access.log

Print all requests to /admin/login.

sed-n'/\/admin\/login/p'access.log

Print all POST requests.

sed-n'/^POST/p'access.log

Print requests containing ../.

sed-n'/\.\.\//p'access.log

Print requests containing <script>.

sed -n '/<script>/p' access.log

Print requests containing OR.

sed-n'/\bOR\b/p'access.log

Print all requests except successful requests.

sed -n '/200/!p' access.log

Print all requests from 10.0.0.25.

sed-n'/10\.0\.0\.25/p'access.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/200/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/401/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/403/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/\/admin\/login/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/^POST /p' access.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/^POST/p' access.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/\.\.\//p' access.log
10.0.0.30 - GET /../../etc/passwd 403
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/\bOR\b/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/200/!p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/10\.0\.0\.25/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

Exercise 7: Labelling Suspicious Web Requests

Use substitution to label the events:

Replace status code 200 with [ALLOWED].

sed 's/200/[ALLOWED]/g' access.log

Replace status code 401 with [UNAUTHORIZED].

sed's/401 / [UNAUTHORIZED]/g'access.log

Replace status code 403 with [BLOCKED].

sed's/403 / [BLOCKED]/g'access.log

Replace /admin/login with /RESTRICTED_RESOURCE.

sed's/\/admin\/login/\/RESTRICTED_RESOURCE/g' access.log

Replace GET with READ_REQUEST.

sed's/GET/READ_REQUEST/'access.log

Replace POST with SUBMIT_REQUEST.

sed's/POST/SUBMIT_REQUEST/'access.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/200/[ALLOWED]/g' access.log
192.168.1.10 - GET /index.html [ALLOWED]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html [ALLOWED]
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html [ALLOWED]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/401/[UNAUTHORIZED]/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/403/[BLOCKED]/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200
```

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/\/admin\/login/\/RESTRICTED_RESOURCE/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/GET/READ_REQUEST/' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 's/POST/SUBMIT_REQUEST/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$
```

Exercise 8: Grouping Security Operations

Use { } to perform multiple commands.

For entries containing status code 401

Replace 401 with [UNAUTHORIZED].

Print the modified line.

sed'/401/{s/401/[UNAUTHORIZED]/; p;}' access.log

For entries containing status code 403:

Replace 403 with [BLOCKED].

Replace the source IP address with [REDACTED_IP].

Print the modified entry.

sed'/403/{s/403/[BLOCKED]/;s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/; p;}' access.log

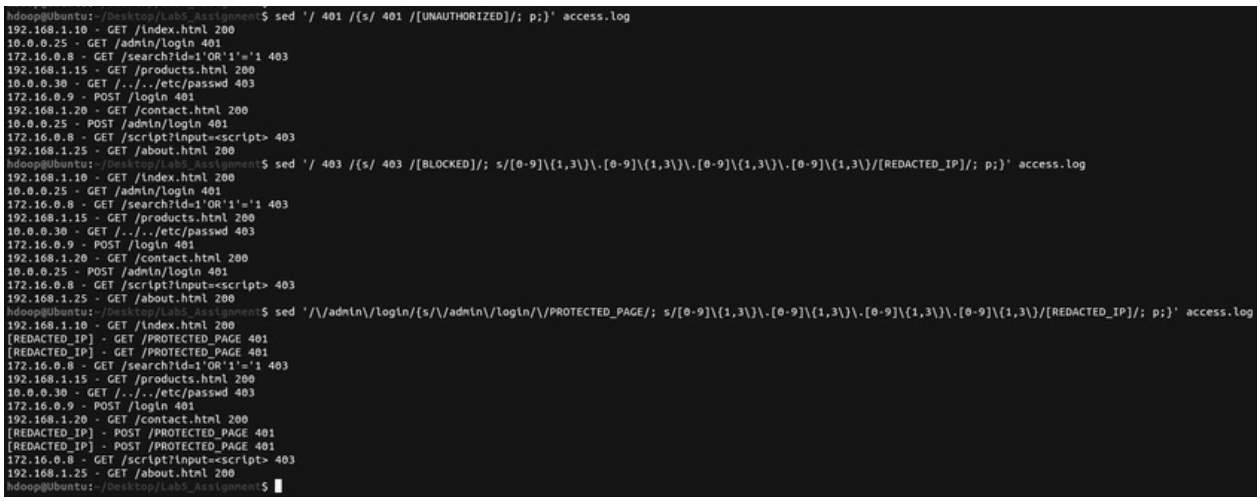
For requests to /admin/login:

Replace the URL with /PROTECTED_PAGE.

Replace the source IP with [REDACTED_IP].

Print the result.

sed'/\/admin\/login/{s/\/admin\/login/\/PROTECTED_PAGE/; s/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/; p;}' access.log



Exercise 9: Adding Incident Labels

Insert the following heading before the first line of auth.log:
SECURITY AUTHENTICATION LOG

sed '1i\SECURITY AUTHENTICATION LOG' auth.log

Append the following message after the final entry:
END OF SECURITY LOG

sed '$a\END OF SECURITY LOG' auth.log

Insert [WARNING] before every failed login.

sed'/Failedpassword/s/^/[WARNING]/'auth.log

Append Review this event after every invalid-user entry.

sed'/Invaliduser/s/$/ - Reviewthisevent/'auth.log

Insert [SUSPICIOUS WEB REQUEST] before every 403 entry.

sed '/403/s/^/[SUSPICIOUS WEB REQUEST] /' access.log

Append [REQUIRES INVESTIGATION] after every 401 entry.

sed '/401/s/$/ [REQUIRES INVESTIGATION]/' access.log

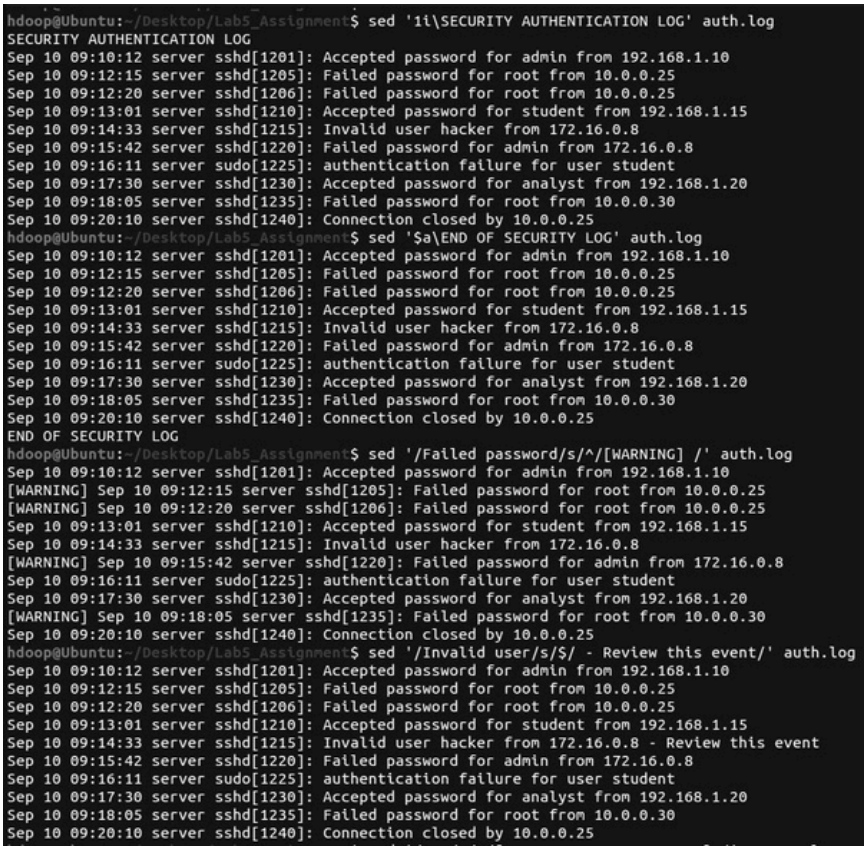
Replace all entries containing Connection closed with:
[CONNECTION TERMINATED]

sed '/Connection closed/c\[CONNECTION TERMINATED]' auth.log

Replace all successful web requests with:
[NORMAL WEB REQUEST]

sed '/200/c\[NORMAL WEB REQUEST]' access.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/403/s/^/[SUSPICIOUS WEB REQUEST] /' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
[SUSPICIOUS WEB REQUEST] 172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
[SUSPICIOUS WEB REQUEST] 10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
[SUSPICIOUS WEB REQUEST] 172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/401/s/$/[REQUIRES INVESTIGATION]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401[REQUIRES INVESTIGATION]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401[REQUIRES INVESTIGATION]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401[REQUIRES INVESTIGATION]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/Connection closed/c\[CONNECTION TERMINATED]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/200/c\[NORMAL WEB REQUEST]' access.log
[NORMAL WEB REQUEST]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
[NORMAL WEB REQUEST]
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
[NORMAL WEB REQUEST]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
```



Exercise 10: Separating Log Entries with G

Use G to:

Add a blank line after every authentication event.

sed 'G' auth.log

Add a blank line after every failed login.

sed '/Failed password/G' auth.log

Add a blank line after every blocked web request.

sed '/403/G' access.log

Add a blank line after every second entry starting with the first entry.

sed '1~2G' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15

Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25

Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '/403/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403

192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403

172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403

192.168.1.25 - GET /about.html 200
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed '1~2G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25

Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student

Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$
```

Write failed logins to failed_logins.txt.

sed-n'/Failedpassword/wfailed_logins.txt' auth.log

Write successful logins to successful_logins.txt.

sed-n'/Acceptedpassword/wsuccessful_logins.txt' auth.log

Write invalid-user events to invalid_users.txt.

sed-n'/Invaliduser/winvalid_users.txt'auth.log

Write 401 web events to unauthorized_requests.txt.

sed-n'/401/wunauthorized_requests.txt'access.log

Write 403 web events to blocked_requests.txt.

sed-n'/403/wblocked_requests.txt'access.log

Write all events related to 172.16.0.8 to suspicious_ip.txt.

sed -n '/172\.16\.0\.8/w suspicious_ip.txt' auth.log

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Failed password/w failed_logins.txt' auth.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Accepted password/w successful_logins.txt' auth.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Invalid user/w invalid_users.txt' auth.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/401/w unauthorized_requests.txt' access.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/403/w blocked_requests.txt' access.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/172\.16\.0\.8/w suspicious_ip.txt' auth.log
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ ls
access.log  auth.log  blocked_requests.txt  failed_logins.txt  invalid_users.txt  successful_logins.txt  suspicious_ip.txt  unauthorized_requests.txt
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$
```

Label successful and failed logins.

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN] Accepted password/' -e 's/Failed password/[FAILED LOGIN] Failed password/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Label 401 and 403 web requests.

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -e 's/401/ [UNAUTHORIZED] 401 /' -e 's/403/ [BLOCKED] 403 /' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login  [UNAUTHORIZED] 401
172.16.0.8 - GET /search?id=1'OR'1'='1  [BLOCKED] 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd  [BLOCKED] 403
172.16.0.9 - POST /login  [UNAUTHORIZED] 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login  [UNAUTHORIZED] 401
172.16.0.8 - GET /script?input=<script>  [BLOCKED] 403
192.168.1.25 - GET /about.html 200
```

Delete successful logins and print failed logins.

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -n '/Accepted password/!{/Failed password/p}' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

Replace root with [PRIVILEGED_USER] and mask all IP addresses.

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -e 's/\broot\b/[PRIVILEGED_USER]/g' -e 's/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
```

Replace GET and POST with READ_REQUEST and SUBMIT_REQUEST.

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -e 's/\bGET\b/READ_REQUEST/g' -e 's/\bPOST\b/SUBMIT_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST /../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

Create security_report.sed containing commands to:

Insert SECURITY INCIDENT REPORT before the first line.

Replace Failed password with [FAILED LOGIN].

Replace Invalid user with [INVALID USER ATTEMPT].

Replace Accepted password with [SUCCESSFUL LOGIN].

Replace all sample IP addresses with [REDACTED_IP].

Append END OF INCIDENT REPORT after the last line.

Execute it using:

sed -f security_report.sed auth.log

Save the output:

sed-fsecurity_report.sed auth.log > incident_report.txt

```
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ gedit security_report.sed
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -f security_report.sed auth.log
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
hdoop@Ubuntu:~/Desktop/Lab5_Assignment$ sed -f security_report.sed auth.log > incident_report.txt
```

Modifying a Security Configuration File

Create security.conf:

PermitRootLogin yes

PasswordAuthentication yes

MaxAuthTries 6

FirewallStatus disabled

LogLevel INFO

GuestLogin enabled

First make a backup:

cpsecurity.confsecurity.conf.backup

Use sed -i to:

Change PermitRootLogin yes to PermitRootLogin no.

Change MaxAuthTries 6 to MaxAuthTries 3.

Change FirewallStatus disabled to FirewallStatus enabled.

Change LogLevel INFO to LogLevel VERBOSE.

Change GuestLogin enabled to GuestLogin disabled.

Use the safer backup form:

sed -i.bak 'sed-command' security.conf

This exercise demonstrates configuration editing only; it does not configure the actual system.

Write a Bash script named security_log_analyser.sh.

The script should:

Accept a log filename as a command-line argument.

Check whether the argument is provided.

Check whether the file exists.

Extract failed login attempts.

Extract invalid-user attempts.

Replace sensitive IP addresses with [REDACTED_IP].

Add the heading SECURITY ALERT REPORT.

Add [ALERT] before suspicious entries.

Append END OF REPORT.

Save the result in security_alert_report.txt.

Display the generated report.

: cp security.conf security.conf.backup

sed -i.bak 's/PermitRootLogin yes/PermitRootLogin no/' security.conf

sed -i.bak 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf

sed -i.bak 's/FirewallStatus disabled/FirewallStatus enabled/' security.conf

sed -i.bak 's/LogLevel INFO/LogLevel VERBOSE/' security.conf

sed -i.bak 's/GuestLogin enabled/GuestLogin disabled/' security.conf

Bash Script:

#!/bin/bash

Accept a log filename as a command-line argument and check availability

if [-z "$1"]; then

echo "Error: No log filename provided."

echo "Usage: $0 <log_filename>"

exit 1

fi

LOG_FILE="$1"

Check whether the file exists

```
    if [ ! -f "$LOG_FILE" ]; then
    echo "Error: File '$LOG_FILE' does not exist."
exit 1
    fi
    OUTPUT_FILE="security_alert_report.txt"
    sed -e '1i\SECURITY ALERT REPORT' \
-e '/Failed password/s/^/[ALERT] /' \
-e '/Invalid user/s/^/[ALERT] /' \
-e 's/[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}/[REDACTED_IP]/g' \
-e '$a\END OF REPORT' \
"$LOG_FILE" > "$OUTPUT_FILE"
    echo "Report generated successfully: $OUTPUT_FILE"
    echo "----------------------------------------"
# Display the generated report
cat "$OUTPUT_FILE"
```